from Bacon to Machiavelli. In Shakespeare, a century after Poggio, atoms make a delightful appearance:

> MERCUTIO O, then I see Queen Mab hath been with you:
> She is the fairies' midwife, and she comes
> In shape no bigger than an agate-stone
> On the forefinger of an alderman,
> Drawn with a little team of atomies
> Over men's noses as they lie asleep …[22]

Montaigne's *Essays* include at least a hundred quotations from Lucretius. But the direct influence of Lucretius extended to Newton, Dalton, Spinoza, Darwin – and all the way up to Einstein. The very idea of Einstein's that the existence of atoms is revealed by the Brownian motion of minute particles immersed in a fluid may be traced back to Lucretius. Here is a passage in which Lucretius provides a 'living proof' of the notion of atoms:

> This process is illustrated by an image of it that is continually taking place before our very eyes. Observe what happens when sunbeams are admitted into a building and shed light on its shadowy places. You will see a multitude of tiny particles mingling in a multitude of ways in the empty space within the light of the beam, as though contending in everlasting conflict, rushing into battle rank upon rank with never a moment's pause in a rapid sequence of unions and disunions. From this you may picture what it is for the atoms to be perpetually tossed about in the illimitable void. To some extent a small thing may afford an illustration and an imperfect image of great things. Besides, there is a further reason why you should give your mind to these particles that are seen dancing in a sunbeam: their dancing is an actual indication of underlying movements of matter that are hidden from our sight. There you will see many particles under the impact of invisible blows, changing their course and driven back upon their tracks, this way and that, in all directions. You must understand that they all derive this restlessness from the atoms. It originates with the atoms, which move of themselves. Then those small compound bodies that are least removed from the impetus of the atoms are set in motion by the impact of their invisible blows and in turn cannon against slightly larger bodies. So the movement mounts up from the atoms and gradually emerges to the level of our senses, so that those bodies are in motion that we see in sunbeams, moved by blows that remain invisible.[23]

Einstein resuscitated the 'living proof' presented by Lucretius, and probably first conceived of by Democritus, and made it solid by translating it into mathematical terms, thus managing to calculate the size of the atoms.

The Catholic Church attempted to stop Lucretius: in the Florentine Synod of December 1516 it prohibited the reading of Lucretius in schools. In 1551, the Council of Trent banned his work. But it was too late. An entire vision of the world which had been swept away by medieval Christian fundamentalism was re-emerging in a Europe which had reopened its eyes. It was not just the rationalism, atheism and materialism of Lucretius that were being proposed in Europe. It was not merely a luminous and serene meditation on the beauty of the world. It was much more: it was an articulate and complex structure of thinking about reality, a new mode of thinking, radically different from what had been for centuries the mind-set of the Middle Ages.[24]